A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \ldots, b_{M-1})$, $B_N = \sum_{n<N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1}; "rep bits" is the bit length of its least positive representative r_N, "mod bits" is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1.

#	mean b	$\|D_3\|$	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997

#	mean b	$\|D_3\|$	$P(16)$	B_{240}	rep bits	mod bits	fill
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000

#	mean b	$\|D_3\|$	$P(16)$	B_{240}	rep bits	mod bits	fill
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

3 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$\lvert D_3 \rvert$	B_{240}	rep bits	mod bits	fill	rep bits at N=100/200
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

```
122121221212212121221212212121221212212121221212212122121212
212122121212212122121212212121221212212121221212212122121221
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$\theta = 0.10$:

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122121212212122121221212122121221212121221212212121212121221212
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122121221212212121221212212121221212212121221212212122121212
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$\theta = 0.20$:

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121221212212121221212212212122121221212122121212121212212
```

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122121212212122121221212122121221212121221212212121212121221212
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$\theta = 0.33$:

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121221212122121221212212122121221212122121212121212212122121212212
121221212212121221212212212121221212212121212121212212
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4 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_N+1} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x+1)/2^{v_2(3x+1)}$ and a period-p word w with sum B satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \qquad A_w = \sum_{j=0}^{p-1} 3^{\,p-1-j}\, 2^{B_j}, \quad B_j = b_1 + \cdots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

$x = -1$: word (1), period 1

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \qquad v_2 = 1, \qquad S(-1) = \tfrac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every k.

$x = -5$: word $(1, 2)$, period 2

$$\begin{aligned} &3(-5) + 1 = -14 = -2^1 \cdot 7, && v_2 = 1, && S(-5) = -7, \\ &3(-7) + 1 = -20 = -2^2 \cdot 5, && v_2 = 2, && S(-7) = -5. \end{aligned}$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \qquad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

$x = -17$: word $(1, 1, 1, 2, 1, 1, 4)$, period 7

$$\begin{aligned} &3(-17) + 1 = -50 = -2^1 \cdot 25, && S\colon -17 \mapsto -25, \\ &3(-25) + 1 = -74 = -2^1 \cdot 37, && -25 \mapsto -37, \\ &3(-37) + 1 = -110 = -2^1 \cdot 55, && -37 \mapsto -55, \\ &3(-55) + 1 = -164 = -2^2 \cdot 41, && -55 \mapsto -41, \\ &3(-41) + 1 = -122 = -2^1 \cdot 61, && -41 \mapsto -61, \\ &3(-61) + 1 = -182 = -2^1 \cdot 91, && -61 \mapsto -91, \\ &3(-91) + 1 = -272 = -2^4 \cdot 17, && -91 \mapsto -17. \end{aligned}$$

Cycle formula: $p = 7$, $B = 1{+}1{+}1{+}2{+}1{+}1{+}4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \ldots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$A_w = 3^6{\cdot}1 + 3^5{\cdot}2 + 3^4{\cdot}4 + 3^3{\cdot}8 + 3^2{\cdot}32 + 3{\cdot}64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \qquad (139 \cdot 17 = 2363).$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the -17 cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \le Cp^{\tau+1}$) governs.

5 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can't prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it's odd, triple it and add one; if it's even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number's entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you'd have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can't do that: probability talks about most numbers, never about that one. It's the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove $\sqrt{2}$ does — even though $\sqrt{2}$ is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance

— never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|$=0.214):
111411211121121111112114421114112222111122212112122212112221
112221122411212122222121212111211111121221222111411112211121
222111111112422222211111212114112121112142211141421211111141
211211412222121111111122222121141212112214111211111114422241

#2 (mean 1.579, $|D_3|$=0.214):
222222222222222111111211211421111211421111221222222111424121 1
111112412111211422111141112111122112212212111111121211111211
212222111212111114221424112111212411211144212111114121114221
211211111111122121111121211221121211141121141242222122222111

#3 (mean 1.579, $|D_3|$=0.214):
222112111112112421221112112111121212211121111211212211444 12
422222211122121211111112111111121141212111222422221411121211 1
112411112112224211112111211122121122111124112111224111121221
224122114112111122122412112421122212111441121112112121111122

#4 (mean 1.583, $|D_3|$=0.239):
111111111111111111111111111111111112122211411421412211142 12
111421141112214212421122114411224111121112212122112222221111
111112221221121111111111121421121124112212121142244122121211
121411412211121111211221111422424421212112121142112221121111

#5 (mean 1.592, $|D_3|$=0.220):
111111111111111111111111111121121111412414211121212411112242
121121121121211114211141222111242122111211222111111111122111
221222112412112111422412112222244212121411111124111221221212
212211211411111122112142111122211122224114212211221214211224

#6 (mean 1.579, $|D_3|$=0.215):
222222224222222421121111141242121141122122222111112112241221
111121121111111111112211111222224211142122142111111112422221
111112221111121111222214111222211212112421111114111121211222
111112111112111112211221214211141122121141214111121211212112

#7 (mean 1.592, $|D_3|$=0.215):
121112111211212121111224121112121411212114112411221242222211
221221111112211111221212224214111211122112111122211121111211 2
212422141111221412421111222214112142112111121211111122111111
111141212222242111111221211221111212421212114112211222141211

#8 (mean 1.579, $|D_3|$=0.216):
122111111111212112111114112122221111121111211111121114221141 4
222221141211111421222112121111112114122121211122121211421121
412121124112211112422112221111111411421211122222411112111221
214122121114112111211221141224111112112112412121221211222221

#9 (mean 1.579, $|D_3|$=0.229):
222222222222224222222222222222222222222422211122211221144111
121121122111221112211111211142124111122211121121111222411111
241212121141111211121111212211111221112111141212121142111111
111122112211111211111142242121121121111221111111111112112421

#10 (mean 1.579, $|D_3|$=0.233):
111112111222121222121211114111414222441214211212124111122211
111111111121121421111411122221211224211422141122144142111121
142141111212211111212222411111111111111224221212122111111111
111221422111212111111111142211121111212111211211121114122221

#11 (mean 1.579, $|D_3|$=0.229):
2222224222222422222221122112212412212211122111422111112122112
1111121111211212111121112211211211221122111122241121211241141
121211114211121121122111412121211214222111112112411211222121
224221212212212112111421211121412211111112211111111212211111

#12 (mean 1.579, $|D_3|$=0.216):
422211212114121122244111122221112411221221112121112441124421
112222111112111211122121112212212212111111111121242212112122
114122214121111111241241111211411212111111141112221141211212
2111121111211111111222421121114211111212212121121111212211221

#13 (mean 1.592, $|D_3|$=0.233):
111111111111111111111111111111111111111111111111222212111121
122124214112211111112211111222122141114421122211212442114111
121241421122224122111422121211222421122114421142422121221111
21112414421212211212121222111211124111211112121114212122141

#14 (mean 1.579, $|D_3|$=0.215):
242222222242222222211111112112211111111111111111221121224211
211121212222112122121412112122112122121121212111121142111121
212111221114111121121244122211211122111122124124112112222212
211211111111211214114111112124111224141124122121211111211411

#15 (mean 1.592, $|D_3|$=0.216):
11111111111111111111111111111111111244111221112111112112211221
121221121141241214211241221111111224121122222211111222211122
122212222142121212142222241121411121121112221411414111122111
211211221121242112114124211211224114212222211141122222111111

#16 (mean 1.583, $|D_3|$=0.214):
2122141111121111212111112122111212111121122414224112111221122
241122142121211111124121124441122122111111122212124222211214
211121421111112122111124211221211412111111221122111111121222
114121211411211111124122212122121111212121221212112111211122

#17 (mean 1.592, $|D_3|$=0.215):
111211212242121222211221222111121224211121121122222114111212
211111242122412112112121112211141122141121124221112221211411
411111421211221112112211112211121111111221221121221121114112
212221111111114111411211214111111222112124211121122242421222

#18 (mean 1.579, $|D_3|$=0.218):
222222224242211124111122122121141122122112121221122114241112
111224211221112121114111121211411111222121121212412111121222
12114121111222221112111112211412414112211221111212212221111
212214221121121111212121111112221114211211122111121111122111

#19 (mean 1.592, $|D_3|$=0.220):
112411111211112121112411212111212211211111412211111211112221
121212111212212112121114111111112141122122442121122112222112
211414211141112112212244111111211211112112221412411141212222
111121122124411212111122121241121241121214242212112211111111

#20 (mean 1.579, $|D_3|$=0.260):
22222242222222222222222222222222222212122111112211111221124112
211112112211224121114111122211212212112221111111412122212212
421114212124221212112112211111121122111112211122124221111211
1212111111212221211112111111111111141121112411212221222121111

#21 (mean 1.579, $|D_3|$=0.272):
222222222222222222222222222222222222121211112212211121122222121
1111112212112111122211112111112111112221112122111141121414121
111211211121422211211212121122421212211112414212111121121121
1211211121111111122121211221121122212222124412212111111412122

#22 (mean 1.592, $|D_3|$=0.220):
11111111111111111111111111111211112242124211142124122111112241
221114214212211122221221414421121112221121141222111211112212
114111112111411212111121111224411212111221121222421112111114
121111121211212124242212121241212211122411111212122122211212

#23 (mean 1.587, $|D_3|$=0.228):
142222112411411141112111121121122414422122111111211211222412
222121111211221211111211122114112121214111111211111221122122
124221212111111412142112112221111212111114121112421211112112
1224111111111111111211111222221111224244111241142141111122111

#24 (mean 1.587, $|D_3|$=0.246):
111111111111111111111121142122111111111111111111122122211412
12121212241421141111111214212211411121124212122222414114212
211414122212111221211121112111241111111221142111211112122141
212111142121111214124412121221222112221411121421141114211112

#25 (mean 1.592, $|D_3|$=0.222):
1111111111111111111111111111111242212211112222112242111111411
121112221221112421142211241212122114111121122124112221241421

2211124122222212122111222242242112121242112142222212111111122
2111111212112211221221211122422212111222211111111221221122 11

#26 (mean 1.592, $|D_3|$=0.242):
1111111111111111111111111111111111111111111111111111111121111111
2212222211111221422411122421111141111112211121442224211211 24
2142441211211221121411212122414114212211114111222111221412 11
1121122224421122122124122111111141242221221211412422122112 11

#27 (mean 1.579, $|D_3|$=0.222):
2242222222222222222222211112221111212111211112111111112111 1111
1222112212422221221211111111221411121222121111142111214142 2
4111211211111111212112224114112211222111111111221112121121 12
2214241222224212211111111111411112142222111111211422212121 121

#28 (mean 1.592, $|D_3|$=0.214):
1111111111111111111111111111111211144112141112121112111121122
4112141121112221221122122122222224221224121112122121124121 11
1121121214212112111411122111242112411224212222211221121221 21
1212111222111121121442412111224422211212111122121114122112 12

#29 (mean 1.587, $|D_3|$=0.215):
1121124142111111112124221111111111112111222111411211244211 212
4121121212221121222212112212111122111142112212222141212121 124
1112111124221121112424121112111222111112222121112212111121 112
1122121212111111442114414211211212112114211221212121222111 111

#30 (mean 1.579, $|D_3|$=0.260):
2222222222222222422222222222222222222222222141111211211111122 11
2211111112111112411122212112241421221212211411212112121211 22
1111111122221112111211111111111121111111111442221212121414 1221
2122112112111221111111121212111211222241221211111222112222 11

#31 (mean 1.592, $|D_3|$=0.214):
1111111111111111111111111121122112211122114211122122222221 214
2122221111121212111222121122122222111111122122214124121121 111
1411211121111411142114422242112222412111114121112114411222 21
1122214212111111211122112421112241111221224122112222121421 111

#32 (mean 1.583, $|D_3|$=0.239):
1111111111111111111111111111111111111111121111241122114212 14211
2411222421121141442112212111121112211212111212221122121221 12
1122111121111141211421421221142242212112141112212121114122 12
2211111121412211121114114441114212212411221111211122212241 11

#33 (mean 1.579, $|D_3|$=0.218):
4212412221211211211121211212221111212411111111222121121122 41
2221111211111121412211242221221121112114111112221122111211 11
1111112122111111221112211222212441111412421211221122142211 12
1212212112222121112211122111142122222111124222111111111244 12

#34 (mean 1.587, $|D_3|$=0.217):
1141111211111111411112144142414111211211112211112221112211 1122
1121111222222211221122121212211221122111111111211112421122 1211
1124121221212222241224212111111112141112212422112112122114 12
2112212112121112111411111111111111411421411214222211121421 121

#35 (mean 1.592, $|D_3|$=0.214):
1111111111111111141222111211121214214421221212422121212412 14
1121121224211221111214112412222122421112211212222411112121 22
1111111212112121412122121124124122221111114112211121222412 21
2221111221111112111211121122412112221111121241111211121111 11

#36 (mean 1.592, $|D_3|$=0.216):
1111111111111212122221212111112121221142214112122222121121 211
1121121122121111112114121142121112211211411111114111121124 211
1111421122114221221141114112242111121241412214222211242141 21
2124111111221121211112114111121221221222111224112122112122 11

#37 (mean 1.579, $|D_3|$=0.220):
2222222411111111121211244221212111111211111111114121212411 224
1242111112111221212121114221111112121111111111122112414112 212
1122112142111212242111221211111111112124124111111424112111 11
2221211122421122412211112222124111121211122412112211112111 21

#38 (mean 1.592, $|D_3|$=0.225):
1111111111111242112112121122212212111124112121111211124111 12
1212221224422112221142422112111112222121141111211221211122 142
1412144421111241242241211212111214212141121111221111111241 121
2211111411121112111211112111111122112211422211111111212141 12

#39 (mean 1.587, $|D_3|$=0.221):
2241112122211411111122412221114112111111114112121112412112 11
1112111222212141111422114112211221122121112122121111121111 11
2414112111212421212111422221111141211211211112121211211111 11
1442212111121241221221112222111211211142212121111224442112 211

#40 (mean 1.579, $|D_3|$=0.218):
2222222222221141122111111214111211111211111211121112224112211 1
11212141111122411121111222212211221112211211221211141214221 2
22212212222122211211112111411222112121111214111122112221141 1
12111121121241141221221221212121141122144121111111124211121 2

#41 (mean 1.592, $|D_3|$=0.225):
111111111111111111111111111111111111122112111111221412214122 11
2111122211121412211221111222111111122111211214212414242412 11
4221211111411111141111241112121212221112221222141222212212 12
2421212241114214422121111122211121241111412122421111124122 12

#42 (mean 1.579, $|D_3|$=0.238):
22222222222222222222222222222111221114112111111114112412111 1
121241112411122112121111112111211211111222111121122121121 21
211114121111112211112212212121112121112112121111112122111 1
2122411124111221222122111122111242221141121212122422222124 41

#43 (mean 1.579, $|D_3|$=0.215):
2224222422211212242112141412112111244112121111121111241112 22
111112221111222412211111111112121211421121111212112224214 221
2211112121111221221121211112111121221111222122112111111222 14
2122412112111112212121111211211422111111142111112122214121 21

#44 (mean 1.592, $|D_3|$=0.215):
111111111111111112111221221122111114211122111122421112121 1411
42221111221111122244211111221111212241211121222141221142221 1
11221421122111121111221224112121122122112112212111211112121 4
1121121111214111122221141222211212122212244114422212221121 12

#45 (mean 1.587, $|D_3|$=0.217):
1211211224212211141411111111112121224112211111111111211112111
1221122222112112122241112411412222412421412112211212214214 12
1111111222122121114221111212211121121141212111122222222114 12
11111111221212112111112121411124111114112122211214221111114 1

#46 (mean 1.579, $|D_3|$=0.215):
2222422222222222222222221211111211111211214221114411221111112
1241111111121211111211112222111111212212112111112211111224122114
211121144411111421221112222111111211111212212112111222221111 1
2141121412412141121142111121212112111211421121111121111121212

#47 (mean 1.579, $|D_3|$=0.229):
2242222222222222222224222442222222122111124111144114211111111
1111212112111112414121111121221212111122211212112112221112 11
11141112121112211111122121222211221121211211111111212211111
1214211211222122421112222122221212211111111111211121211214 11

#48 (mean 1.579, $|D_3|$=0.215):
2422222224212211121111111112121112441141111111224221214211 12
21112122221222124111211221221221211212122121242111114111121 1
12141411111111111211141121121222214212111212211212221211122 1
2424212212111111111111122221111222112111141121111121122121 12

#49 (mean 1.592, $|D_3|$=0.215):
1111111111411211422221221144112112214212142421211212442111 11
21222112112112122142122114112121112122121211222222121211121 1
22111112112211111111211212411121111221111122121211121211111 22
2222111121141212222114421421222111112411112222412121211111 11

#50 (mean 1.592, $|D_3|$=0.216):
12211212142111124212112211212121122121222142142111112111112 1
14122111114142111122212212141212211111441121111111111111212 21
21114412111211121121221122112222121141212111111112111411121 2
1214112421242122121221221241122111141211121121224211112212 11

#51 (mean 1.579, $|D_3|$=0.229):
2222222221221242111224122211221121121112211222141221111121 12
2111244112111214211121114111122111122244211111111211222111 21
2121221211111222222111221211412212221121221112112111221122 21
1111121222211411141141121211111214112221211212211221112121 12

#52 (mean 1.579, $|D_3|$=0.248):
222422222222222222222222222222222222222212211121111221111121 21
2111222111111121111241111122221121111111222111121121122111211
2421111111122112122421222112442112414121111221112211121111111
411112212211121211111124112211211212114411211121121121221121

#53 (mean 1.592, $|D_3|$=0.218):
11111111111111111142121114422111211121211211421221411111121111
4121221214222222221211112121121222222141111221212111141122212
2144111142411111214221112212121112121112121212411112221112 12
111212212112222112122422221212211411121111221221122212221112 11

#54 (mean 1.579, $|D_3|$=0.215):
242222222442222222442222222222224211111212221111111211111111111 42
21121111212222111212111114211222114111112112212211211111221 11

211242111221211111121211142121111141212121121111211111221122
111144221221211142211112112111122211111221121114121111211421

#55 (mean 1.592, $|D_3|$=0.233):
111111111111111121144111141412142211212111221111212211121111 1
221121211111222121111112211211214211122412111122212211141112
211112142112121241112214224112411221241211112112112111212112
221141111242241212141211111411211424222121122114114111121121

#56 (mean 1.592, $|D_3|$=0.279):
111111111111111111111111111111111111111111111111141112421241 12
241122111111211211421111141212242211111221212442111121111141
442122221411121121122112142112422411114211214211212411112411
211111122212122122141211111221214412211212211441411111211142

#57 (mean 1.579, $|D_3|$=0.215):
222222222211214112211211221111111211141122212224211211121111
111211421111222221121241222122141121121241111112111211241222
214122112112141111111112142112421111122121411121222141211211
122112111221422121111142222124211221121121112111212111111111

#58 (mean 1.579, $|D_3|$=0.218):
222222222222211111212111444121114121122121211112112211241244
122111411112112111211211111111212221214212211112212111441112
221121111221112422112112241222121111111114212111242111241211 2
121121112212111211111221212112221122111221211121211221122211 11

#59 (mean 1.579, $|D_3|$=0.238):
222222222222222222222222222222222222222224222221122111111122 11
112112122112112144411211211114121241112112212111211212121111
112421114211422211211122122111112112411121211412111111121212
111112122111111211111421111221112111211111114211211121121211

#60 (mean 1.579, $|D_3|$=0.214):
222422222222222221122111242111211112412122111421111211211141221
111112112121411221111111112241211114212141141111122111141112
111211212211414112111441221212111212112211124212221111122221
111242112112111112121111211412111111221111122211221222221211

#61 (mean 1.579, $|D_3|$=0.238):
222222222222222242211111111121141111212112221114242121111111
121122112124211414221111222221112211111111212211111112112221
212221421211112212221122122111121111111112214221112211112222
212122221211211121142212212144122111211211122111421121212112

#62 (mean 1.583, $|D_3|$=0.230):
111111111111111111111111111112412211111211421211224214141121 2
111221222122141111121111422211412114111411212111121211212241
112211412411112111224211412121121212224412242111141212111411
122122111112211241121121111122111222121121211212212221114122

#63 (mean 1.579, $|D_3|$=0.222):
222222222222222222222211412141121122211221111211112241112211
111221111121212214142211122111411211241112112211112142112211
212114214111111121211112111141114214222221112111121221241111
222122411112112112211111111112212111211122212211222121112211

#64 (mean 1.579, $|D_3|$=0.215):
222222222222222222222211111411112141211411111212122412111112
211121121111214211111141222122121111144111111114112111242112
211111412211221141122211121121411142112112112211111221212211
221212112112111121111111411242211221121221112142211121222121 1

#65 (mean 1.579, $|D_3|$=0.222):
222212221121211111141211214111122212111111212112122122122211
111242411242121112211121211111112211121121222222112111212211
112112122112212212211111221121222214112111212114122121211211
142121421114112212222222121112411211211111414221121111144122 1

#66 (mean 1.592, $|D_3|$=0.216):
111111111111111111111111111111111111111111111111212121122111111
221122212412242112211112111211111221222411142221212112224211
124112411211441121121112224214121141111221122222212121212122
222211212114241212442421121112121421211124211242221212112211

#67 (mean 1.592, $|D_3|$=0.215):
111112112111411111122221212111411411121121111121212411412221
122222111212214114112212122221211211411212112411211122112212
214111214111221221112141112111112121121122122412141112221112 1
412111144211212111212412211212122212111212122211222111211222

#68 (mean 1.579, $|D_3|$=0.216):
222222222222224224242411121211112124122114421211211411111122 1
212112121111111244212112212222211212111211121211211141211112
111212111412212111141211121111111221221112211111112112411212
122121111114111111144112211111111224212112124111222111112111

#69 (mean 1.592, $|D_3|$=0.216):
1111111212241111221211114111112212112122241122212222221142111
1111142212112422121411222242422112142421112211112212112222111
1411111121212122112211111212211112211411111124221211121111111
1212411222112141111114111224211211121114121211211211111421212

#70 (mean 1.592, $|D_3|$=0.216):
2111421211224122241112241121121121121121121121121111412112222
1112211121214121221212111111224112121121212111112211214224
1122211212112121211422112221112112114111422211122241111111111
1241221211141211111221212211211121442111111442241211111111112

#71 (mean 1.592, $|D_3|$=0.214):
1111111111111111112211214121421112122112211114122144221122222
1144144124242112221411121121212111111211142122211122212122111
1222242112212111212222124111114112112211121121111212111111222
2112122112142121211112111221121241211211122141212111211111112

#72 (mean 1.579, $|D_3|$=0.214):
2422422222122111114111211112411211211212122112211111221212121
2244111142211221112241111112121211111211212114212121141242111
2121112112221122214122411112111111212214221212112122111122121
1222122121212112111112111214112111121111224111111112114111114

#73 (mean 1.587, $|D_3|$=0.221):
1211411111224111211211242211212122114112221111211111124214211
1222211111111221211121214121114411122121111212111111221212121
1122111111124124212112221221111141211241111111112111121121241
2111222422141111224112112111212221412121222411112211111411241

#74 (mean 1.592, $|D_3|$=0.215):
1111111111111111111121221121214222221141111221212112121122111
11122122114124211222221111221221212112122241211222221142212212
2122212211121211111222421221111111121212121211111211424414 2
2214112111241112114111121111111221111412242212222211411211124

#75 (mean 1.579, $|D_3|$=0.215):
22112411142121212121211122221111112421211212121121141221211
2121114121141111212111121141121241211241111211212212212244 12
2211112222111121221221111112121212222211411121111422221111211
1142112211211212122121111212212412411212111121221121111114121

#76 (mean 1.579, $|D_3|$=0.225):
2222242111221411122111112421111111211121121241111111214111112
42111112111121212111112241211112142121211121111111112221122122
2111221212212111411211112422124211124112211211111111112112414
211141212112111112111121222121414221111211221114412214421112

#77 (mean 1.583, $|D_3|$=0.241):
111111112222111211111211211112221211414122242142222221242142
221122114124221212122112111124121112121211121121111121222221
121122112121142222111422221111241111211211122111111221221122
1212212211242121111212112212212111221221112212111122121212112

#78 (mean 1.592, $|D_3|$=0.215):
111111111112112111212221144221122221222222111112112212111412
112221122212114241411112221211112111411121244111112112222111
2212222121121211121122214412211111111412122221111121222212222
111122412111221112111121121111422122214214111122111114411211

#79 (mean 1.587, $|D_3|$=0.221):
141211221111111421221212111212112222211121111111111411111111124
112114142221221111121122142122221211121122111111114414124222
111111241121411121112211111124211412111142111221211122112112
1241121411122211122121221121112411411121212212211211111222242

#80 (mean 1.587, $|D_3|$=0.217):
1111121442111122111412121121112112222111122121211121221111121
211221111121414212212111422142212111211211121122112121122142
12111221221212222114122221212111211111122124111114211122242
1221122422211124111421212121121124242121211122111211111121112

#81 (mean 1.592, $|D_3|$=0.225):
1111111111111111111111241411112222112141212121111412114211421
112212111111121121121241211111224241111111221142112111212141
22221121112114221112224111242221111114122112122124122211112
111112121112221211224114211124221242122214211111121412211121

#82 (mean 1.583, $|D_3|$=0.214):
2121121242412111121411212112122112112121121242211111111121214
2241112111211244122211221221112222112121122141142214112424421
1242112211121111121122224111121121212111221112111112112212121
1111111112212211121411221112111122211111221122412112212211111

#83 (mean 1.592, $|D_3|$=0.215):
112212111124212112211211112112111114112212121222111111111122
2211112241112222122212221211112122221121121111422421441221212

2411122212212111214141224422211122411112112121122111112111421
4111121211211211414111112212122141121121211111222111212121111

#84 (mean 1.592, $|D_3|$=0.218):
1221111111211411211222121422211422121122211112121221212112111
1212111121121224214111111411212422111121222211112111441122422
1211111111122121122122222122114412211111212221221114111112144
2111212142211421112121112111211411211112221211222222211122214

#85 (mean 1.579, $|D_3|$=0.222):
2121224411222211422212212121121111121111211141411412111111222
1221211111211212212112112112122112114122212112111211222212114
2221242122112111211211241212141111221241111221122121111122122
4412212212121111221212241112121212212122211121141212121111114

#86 (mean 1.579, $|D_3|$=0.238):
2222222222422224222222222222211221111212241212211121242112214
1121111111111112112214211111122111211111121211214122211111424
2112211111122111121112121112141221212121121111221212211221124
2121222211114114242122112121121111122222121211211141111121124

#87 (mean 1.579, $|D_3|$=0.229):
4222222222222222242224222214222211122212121112112111221114124
1112211121111114221112211111121124112111122122411111111211114
1121222111111111111111411112241112112212111111122112112422224
2211222211111121112112414211212211221222422111121212122112214

#88 (mean 1.587, $|D_3|$=0.227):
2421412121211212121412111411122212121111121221121212121222224
1211122211112111111212412111211122122111111112111121121221214
1212421224122121112121222111111221212222112124212121212211224
1122242111122221212442221111121414121111112114221111111241224

#89 (mean 1.579, $|D_3|$=0.218):
4222222222114111211221112121211221111111212111111111221221214
1211222112121221222111124121121114112114211112112121111211144
2412211111221114111112241411212111211112422212222222121222424
2112211112112112211111112221121111212241114214111241222211124

#90 (mean 1.592, $|D_3|$=0.216):
1111111121121211142122121211121221122111212111121422111421114
2111221111244121112222142121111111121211211211422111124211124
1111212122122221212241421122111212121212121211121114114212114
1111211411122121211112222212214114111122441421121112412241114

#91 (mean 1.579, $|D_3|$=0.216):
4222224222212141112121121212141112111141112121112111111211114
1121121212212411211121412212121111111111112112224111211122214
1244122141421144121211211111241221112111112222122121211112114
2212211111124122111111112111221121111122121141211142141212224

#92 (mean 1.583, $|D_3|$=0.225):
1111112122221112212211411211221122421222211111242212111121124
1241112224112112411211221121111111121224122422111111111122124
2111211122121211121121111221211121212224222222222111221212124
1121121122212412424211412111121111112211111212421112412112214

#93 (mean 1.592, $|D_3|$=0.220):
1111111111111111111112212421211111141412122212111121122121114
1122141111212211111122212412112111124121121212122121221121224
4424211221114211241121211112211211121121141112441141111241224
2211121112112221221212114242122421214112121114111221121221114

#94 (mean 1.579, $|D_3|$=0.214):
4422222222222222422412111111111111211122111121124142112211214
1121121221212242121224112111122111221242121121114111211112114
1412421121111121221111222411412111112412211111111111421111124
2221112211111212111214211121222122111121121111214122121122114

#95 (mean 1.579, $|D_3|$=0.238):
2222222222222422222224222222222222222222112111111111122121214
2411111121111121121221112411211211111214212112112121112112214
1112412222111114111221221111114222412112111121211211111111124
2111122122211111411122122211112422241111111112111222412211114

#96 (mean 1.579, $|D_3|$=0.238):
2222222242222222222222121122421111211111212112124121211122124
2211222212111121111412211111111214112211122112121221111221124
1211141221141111221112112211111222112124141211212124221211144
1222121111111211241221111122221211211111112211211124211221214

#97 (mean 1.592, $|D_3|$=0.233):
1111111111111111111112114142221111211221112222111112111114244
2112114214211142121212241114221241112124211142211222122111124
1122112111111121121212221112111141212112222141214111112112214
1214111112221411111122211221111411211412112411212244111124114

#98 (mean 1.579, $|D_3|$=0.229):
2222222222222222222222211114121111122111221222111111112121221
1212111422112111221111412221221111221112111111211121211111122
1112122414111111114122222212111112111111212112112222114111124
1241212222411211211111242241221211421121111221112241121111212212

#99 (mean 1.592, $|D_3|$=0.218):
1111111111111111111141122114142422421111124124111122112111221 2
1221111414112111221111241221211411211111221121244222212224221
1222121222121211111112112222242141222112211122211221121212212
1221112111121222212122221112211112111222214111111211212122211

#100 (mean 1.579, $|D_3|$=0.215):
2222222111144112112422112111412122112222121411121211111121221
2141122121222122212221222211121214211221221112441141121111111
1144111141121114114111111421122112211211211121212121212122211
2121121111111212122121111112221122212212211111211121121411211

#101 (mean 1.579, $|D_3|$=0.218):
2222222222222111121221211112112111421211212111111221141241211
1211141111121121221111111212124112222112221121211112111122111
1422221121121121114111211121212421222222111142422212211421221
2224111211111412141212121111211112111121212112112111424122111

#102 (mean 1.579, $|D_3|$=0.218):
2222222222222222222222222242222222222221212112242112221224121 12
2121211112112121112212211411211111411242111111121111121114111
1211221111121111111211111421121114211111111111122411124442112 1
1122111111111211121121411111211111242411221111112211212112112 1

#103 (mean 1.579, $|D_3|$=0.229):
2222222222121111121112111224112222211212111221221112211111122
1221111212421121121212111212412121211212224421211121214422111
2112121211111112422121121121211242211112211111222111121211112
1214121111211212122112221112412221241211142111111121241121112

#104 (mean 1.587, $|D_3|$=0.217):
1211422221212121221211142111221111221221122214121121212222221
1111214112111211122114111111222121111122111111211111221222222
2114212112111111211211112141114214112122211122111121421241221
2111112141222121141212112221121111142211411221212212224211141

#105 (mean 1.592, $|D_3|$=0.229):
1111111111111111111111111111111212122111112122112211211411221
1121211111111121122111222122242111214211242122211212224222222
1124212121111121222112244111222224212221121111222124221222241
2221122411211212411122121212124211222111211211121212412122122

#106 (mean 1.583, $|D_3|$=0.232):
1111112114222212212121112411111211222241142211221221114221111
2111222212111221122212221121121111222212111212121212422211122
2121221111211221111122121111222111422421121422112211241214 2
1221411141212112112221111112111112242111221211111221112211122

#107 (mean 1.579, $|D_3|$=0.218):
2422222222222122111111111122214121222111111221122111111111111
1212111111111421211221111114222241121211112111111111121212221
2221212221111211121211422111121212114121421121111221212212211
1121111111441221221111422221214122221111242221141222221111124

#108 (mean 1.579, $|D_3|$=0.214):
2422222222222222121221211221112121122122112111122224214414122
1112111111121111141121111222111111122142111122111111142211122
1111111222222222122211414111114142111212111111112112121411212
1111221414441111222111112121112111111124211222211111121121114 1

#109 (mean 1.592, $|D_3|$=0.220):
1111111111111122421242142141122111214212221211221112421112114
2111212111112214114142121112124211111111122122211112421421 21
1121422122221111211111124124221111111121211211111212211211212
4121111112211112111211111111121211142422211412242122221112122

#110 (mean 1.592, $|D_3|$=0.220):
1111111111222142112122212212111111411214211124124112111142121
2111112111111111124112121211111222222124112111114222212211122
2422222111111211421211111111211121122442111121111112121124211
1122122111121111112141112214211421221221242114212141121112111214

#111 (mean 1.579, $|D_3|$=0.229):
2222222121221211211112121121112111111112114212221221111141212
1241121212221211111122121221124111112221221111211211211222111
1224141111111122112444121221212111112112242122221121221122221
1121212211122112121121212124112121122111212421112112214224 21

#112 (mean 1.583, $|D_3|$=0.218):
1111111111121211121141221221412211122211111211141111111122111
2411214212111121212124114221124221111211122221441412142221 21

112121121222111222112111211211111111212211111121121111121121
212442111411122212122112221122211112442422211121111442212111111

#113 (mean 1.579, $|D_3|$=0.248):
222242222222222222221121222111222211121111142211121212111241
111112212121212242111121122211122112112212121111211221111111
112222112211112112122211211111422114212122112141411112211112
212411211212111121122112112214122221121422212121422211111112

#114 (mean 1.579, $|D_3|$=0.229):
222222222222222222111111211211211112121212124111122411112222
112222212211111211111211212111411222114124111111222112411211
211214212111111111114412112221221111212114212411212111221222
142212121122221212221121211212121114111211211211211112214211

#115 (mean 1.592, $|D_3|$=0.220):
111111111111111111211211211142222212212222242212211211111411 1
222212121112111111111144111422121212141112422141242112112112
121211411121221222111111241214422111222122121114121121241211
122211441112112121112111211111121111421111222411422211211111

#116 (mean 1.592, $|D_3|$=0.233):
111111111111111111111111111111111111111111121121121142122212 44
112111422124111121241222121142222121242122112121222111222212
221414412114111111121212421141214214121242122111111211112222
112111111112114121121112411111224222141121222124124111121111

#117 (mean 1.579, $|D_3|$=0.222):
222222221122222111141111122214211221212422112211111111222412 21
211211211211222111211121211112111121111222122421111112121142 2
111111121111212222411111211211221111121112142111111221121222
412222112212141111121412222211111212114124211241222114112221

#118 (mean 1.587, $|D_3|$=0.214):
124122242222141112121111112121211212211111212111211111214221 11
211221111121111111221141111222111112222121111211121111242212111
111221124211122111111411211121221111214211224221112222122414 4
124221222211111142121111211111242411212222211111121121222121142

#119 (mean 1.579, $|D_3|$=0.218):
222242222222222242222222222222222242112112112141111111112211 42
111212111211112121222122112111221211122111211211111111211241 11
111211112211111421111111211111124142122422141111222212111112 2
141121122111111111211212114121122111111211211111111414222412 11

#120 (mean 1.579, $|D_3|$=0.214):
222222422224222222212241121212211111211111111121211211212121 11
212211121121111211211111121141221122111211221112222414111121 11
121122121211122114121121211412211111411112212111111211422444
411211111112121122224114122124124111212112121111111121121211221